

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			award (2) for 6 correct points (tolerance $\pm\frac{1}{2}$ square) award (1) for any 4 or 5 correct points award (1) for straight line through points does not need to be drawn to origin		2	1	3	3	
	(b)			award (2) for high-level quantitative description - as the concentration doubles, the volume of gas doubles - concentration and volume of gas are directly proportional award (1) for lower-level description - as the concentration increases, the volume of gas increases - concentration and volume are proportional - concentration and volume are directly correlated - concentration and volume have a linear relationship			2	2		2
	(c)			more (1) collide (1) gas (1)	2	1		3		1
	(d)			award (1) each for any two of following - increase temperature / warm / heat / hotter - increase surface area (of chalk) / smaller pieces / cut chalk up / powder chalk [do not accept smaller surface area] (add) catalyst (1) award (1) for 'change' surface area <u>and</u> temperature with no reference to 'increase' if no other mark awarded			2	2		2
				Question 7 total	2	3	5	10	3	5